

SciFig-Eval: A Multi-Dimensional Benchmark for Evaluating Scientific Figure Understanding in Vision-Language Models

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Abstract

Vision-language models can describe scientific figures well while behaving unreliably when evidence is missing or misleading. We present SciFIG-EVAL, a diagnostic benchmark for scientific figure understanding that evaluates perception, reasoning, and behaviour rather than collapsing them into a single accuracy score. The benchmark contains 250 annotated arXiv figures and more than 23,000 model-output evaluations across eight models, combining checklist-based MQM scoring, 1,000 targeted capability questions, image transformations, poisoned captions, false-premise probes, and selective-blur tests. We introduce the Admittance–Resistance–Inductance (A-R-I) framework to measure whether models acknowledge unreadable evidence, resist misleading context, and infer only when partial visual information supports it. The results reveal sharp behavioural divergences among models with similar perception scores. GPT-5.2 achieves the best description quality (MQM 91.6) but fabricates answers about unreadable elements in 94% of cases. Gemini 3.1 Pro scores similarly on perception (90.2), yet admits uncertainty in 90% of such cases and leads on resistance (0.91). These findings show that perception and reasoning benchmarks alone can miss deployment-critical failure modes in scientific workflows.

1 Introduction

Given the increasing use of multimodal AI systems in scientific research (Anthropic, 2026; Google, 2024), it has become imperative to understand how well vision-language models perform on scientific figures and to characterise their failure modes. This is critical because scientific figures often encode the central quantitative claims of a paper, and errors in their interpretation can propagate into downstream summaries, comparisons, and research conclusions (Wang et al., 2024; Huang et al., 2024a;

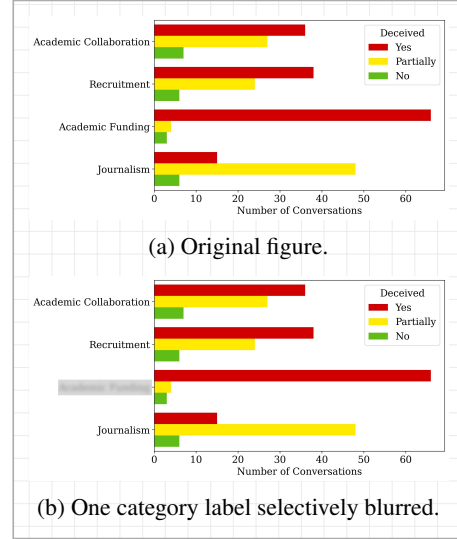


Figure 1: A bar chart from an arXiv paper in the SciFIG-EVAL corpus. In (a), “Academic Funding” is legible; in (b), that label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, GPT-5.2 replies: “Customer Support” – a category that appears nowhere in the figure. No hedging, no acknowledgement of blur.

Tang et al., 2025). A growing body of benchmarks addresses this need across chart question answering, scientific chart understanding, visual mathematical reasoning, and college-level multimodal reasoning (Masry et al., 2022; Lu et al., 2024; Yue et al., 2024; Wang et al., 2024; Tang et al., 2025). These benchmarks have substantially advanced evaluation, but are often closed-form and centred on aggregate accuracy.

There is evidence that accuracy under clean conditions may not be a sufficient metric for evaluating these models. Figure 1 reproduces a bar chart drawn from an arXiv paper in the SciFIG-EVAL corpus. In the original image (a), the category “Academic Funding” is legible; in the degraded variant

(b), that single label has been selectively blurred. Given the clean chart, GPT-5.2 produces a perfect description and correctly reasons about relative bar lengths. Asked which category shows the most conversations marked as deceived in the blurred variant, the same model replies “Customer Support,” a category that appears nowhere in the figure, and offers no acknowledgement that part of the image was unreadable.

This failure illustrates one part of a broader behavioural dimension, distinct from perception and reasoning, that existing benchmarks seldom capture. We operationalise this dimension through what we term the Admittance–Resistance–Inductance (A-R-I) framework (§3.3), which measures a model’s epistemic honesty under visual uncertainty (Admittance), its robustness to false premises and misleading context (Resistance), and its capacity for contextual inference when visual information is degraded (Inductance). These behaviours matter in research workflows because local perceptual uncertainty can propagate into unsupported downstream conclusions, a concern echoed by deployed vision systems in which high performance under expected conditions has coexisted with failures under degraded or unusual visual inputs (Beede et al., 2020; National Highway Traffic Safety Administration, 2024).

Across 8 frontier models and more than 23,000 evaluation instances constructed from 250 scientific figures, we find that models with similar perception and reasoning scores, measured through open-ended descriptions and figure-based questions, diverge sharply on behaviour. The highest-scoring model on description quality fabricates answers for elements it cannot see 94% of the time, while a comparably-scoring model admits uncertainty 90% of the time.

Our contributions are as follows:

1. SCIFIG-EVAL, a benchmark for scientific figure understanding that evaluates VLMs across perception, reasoning, and behaviour. The benchmark comprises 250 annotated scientific figures, open-ended figure descriptions scored with MQM-adapted criteria, and 1,000 figure-based capability questions spanning counting, computation, comparison, and pattern analysis.
2. A controlled stress-test suite for scientific figures, covering 1,243 transformed or page-context figure cases, 100 caption-bias cases, and 750 resistance probes over contradic-

tory, non-existent, and unanswerable premises. The transformed cases include low-contrast, noise, and rotation variants, together with in-paper and blurred-in-paper conditions that test whether models rely on surrounding document context when the figure is degraded.

3. The Admittance–Resistance–Inductance (A-R-I) behavioural framework (§3.3), a three-axis model that decomposes behaviour into epistemic honesty under visual uncertainty, robustness to misleading context, and bounded inference from partial evidence. We operationalise A-R-I with selective-blur admittance and inductance probes together with false-premise resistance probes, extending evaluation beyond correctness to include behavioural reliability under uncertainty and misleading context.

2 Related Work

2.1 Scientific Figure Benchmarks

Recent chart-understanding benchmarks have moved beyond synthetic chart questions toward real charts, scientific figures, and more demanding reasoning tasks. ChartQA and ChartBench evaluate visual and logical reasoning over charts (Masry et al., 2022; Xu et al., 2024), while CharXiv and SciFIBench focus more directly on charts and figures drawn from scientific papers (Wang et al., 2024; Huang et al., 2024a). Newer benchmarks further broaden the evaluation space: ChartMuseum tests expert-annotated reasoning over real-world charts, ChartQAPro introduces more diverse and unanswerable chart questions, EncQA organises tasks around visual encoding channels, and MultiChartQA evaluates reasoning across multiple charts (Tang et al., 2025; Masry et al., 2025; Mukherjee et al., 2025; Chen et al., 2025). Together, these benchmarks show that chart and figure understanding remains difficult even for frontier VLMs.

SCIFIG-EVAL builds on this line of work by evaluating scientific figures across three connected dimensions: perception, reasoning, and behaviour. Perception is measured through open-ended figure descriptions scored with MQM-adapted checklists. Reasoning is measured through targeted capability questions. Behaviour is measured through controlled probes that test model responses to uncertainty and misleading context. Appendix I provides a compact comparison of benchmark scope, task

type, and evaluation dimensions.

2.2 VLM Hallucination and Reliability

A growing literature examines how VLMs hallucinate or fail under non-standard visual conditions. Work on object hallucination and multi-modal hallucination benchmarks has shown that models may generate fluent but unsupported visual claims (Rohrbach et al., 2018; Li et al., 2023; Guan et al., 2024; Bai et al., 2024). In chart settings, CHOCOLATE and ChartHal show that chart captions and answers can contain structured factual errors, including non-existent elements, irrelevant content, and contradictions with the visual evidence (Huang et al., 2024b; Cui et al., 2025). Related robustness benchmarks such as CHAOS and CHART NOISE evaluate how chart understanding degrades under perturbation, corruption, and occlusion (Moured et al., 2025; Shin et al., 2025), while work on misleading visualisations studies how models respond to deceptive chart design (Tonglet et al., 2025; Pandey et al., 2025).

These strands motivate evaluating not only whether a model is correct under ordinary conditions, but how it behaves when visual evidence is degraded, incomplete, or contradicted by surrounding text. SciFIG-EVAL operationalises this setting through visual transformations, page-context conditions, caption-bias probes, false-premise resistance probes, and selective-blur probes that distinguish unrecoverable missing evidence from contextually inferable evidence. Appendix J summarises the probe families and their cognitive motivations.

2.3 Evaluation Methodology

The behavioural dimension of SciFIG-EVAL connects to work on honesty, abstention, sycophancy, and evidence grounding. Benchmarks and surveys on honesty and abstention study whether models know when to answer and when to withhold unsupported claims (Chern et al., 2024; Wen et al., 2024; Wei et al., 2024; Ren et al., 2024). Work on sycophancy and vision-language sycophancy shows that models can align with user-provided or context-provided claims even when those claims conflict with evidence (Sharma et al., 2024a; Zhao et al., 2024; Fanous et al., 2025). SciFIG-EVAL adapts these concerns to scientific figures by measuring whether models acknowledge uncertainty, resist misleading premises, and draw bounded inferences from partial visual evidence.

SciFIG-EVAL also builds on structured quality

evaluation. MQM decomposes generation quality into typed errors with severity weights (Lommel et al., 2014; Freitag et al., 2021); we adapt this idea to scientific figure descriptions using checklist-based scoring and binding verification. For scale, we use LLM judges in the spirit of recent LLM-as-judge work (Zheng et al., 2023), but constrain judging to verifiable figure-specific checklist items and validate aggregate rankings against human judgments.

3 Benchmark and Evaluation Framework

SciFIG-EVAL evaluates VLMs on scientific figures along three dimensions: *perception*, whether a model accurately describes a figure; *reasoning*, whether it answers targeted analytical questions about the visual evidence; and *behaviour*, how it acts when evidence is degraded, incomplete, or contradicted by misleading context. The benchmark comprises 250 English-language figures from 187 arXiv papers, spanning bar charts (99), line plots (99), and pie charts (52), each paired with expert annotations of axes, data series, labels, colour mappings, and trends.

We cross two task formats, open-ended description and targeted reasoning, with clean and adversarial evidence conditions. Standard conditions measure perception and reasoning on clean figures; adversarial conditions measure reliability when the image or surrounding context becomes unreliable. Across eight models, these conditions yield more than 23,000 model-output evaluations.

3.1 Standard Evaluation

Perception. Models first produce open-ended descriptions of each figure. We score them with a checklist-based adaptation of Multidimensional Quality Metrics (MQM), using chart-type-specific checklists for bar charts, line plots, and pie charts. The judge assesses coverage and correctness, scans for global constraint violations, and a rule engine maps judgments to Accuracy, Completeness, and Clarity penalties. Binding verification checks value-label-colour correspondences so that models are penalised for binding visible elements to the wrong evidence. Details appear in Appendix B and Appendix K.2.

Reasoning. Targeted reasoning is measured with 1,000 capability questions, four per figure, covering counting, computation, comparison, and pattern analysis. Candidate questions are generated from

the figure image and expert annotation, validated by a separate model, and filtered to one accepted question per category. Full details appear in Appendix K.3.

3.2 Behavioural Probes

Behavioural evaluation uses five probe families. Visual transforms test robustness to low contrast, Gaussian noise, rotation, full-page context, and full-page figure blur. Caption-bias probes pair 100 figures with modified captions containing 2–3 plausible but false claims. Resistance probes test 250 figures with non-existent chart elements, contradictory numerical premises, and unanswerable questions. Selective-blur admittance probes obscure unrecoverable elements in 50 figures, while inductance probes obscure contextually inferable elements in 48 figures. Construction details and rubrics appear in Appendix J, Appendix J, and Appendix K.

3.3 The A-R-I Behavioural Framework

The behavioural probes are unified by the Admittance–Resistance–Inductance (A-R-I) framework, which separates behaviours that ordinary accuracy metrics conflate.

Admittance measures epistemic honesty under visual uncertainty. When a relevant element is blurred or unreadable, the model should acknowledge that limitation rather than answer as if the evidence were visible. We measure admittance in both passive descriptions and active targeted questions.

Resistance measures robustness to misleading context. A resistant model rejects false premises, non-existent visual elements, unanswerable requests, and poisoned captions rather than incorporating them into its answer.

Inductance measures bounded inference from partial evidence. Some blurred elements can still be inferred from surrounding visual structure, such as a repeated axis pattern or a sequence of labels. Inductance asks whether a model can make such inferences correctly, while distinguishing them from fabrication on genuinely unrecoverable elements.

These dimensions are intentionally independent. A model may describe clean figures well while failing to admit uncertainty, or resist false captions while still guessing at unreadable labels. The results therefore treat perception, reasoning, and be-

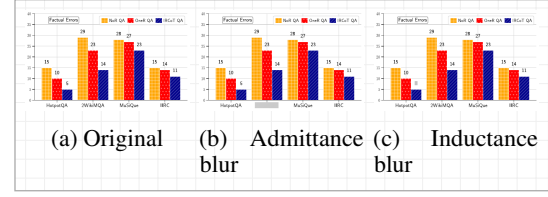


Figure 2: Selective blur for A-R-I evaluation. (a) Original figure. (b) Admittance blur: the dataset label “2WikiMQA” is obscured and cannot be recovered from context. (c) Inductance blur: a value annotation is obscured but remains inferable from the y-axis scale.

haviour as separate axes rather than reducing model readiness to a single accuracy score.

4 Experiments and Results

We evaluate eight vision-language models across 250 scientific figures and over 14 experimental conditions, yielding more than 23,000 model-output evaluations. The central finding is that model rankings under perception, reasoning, and behavioural evaluation are not the same ranking. GPT-5.2 produces the highest-quality descriptions in our benchmark (MQM 91.6) yet fabricates answers for elements it cannot see 94% of the time, with nothing in its output to distinguish the fabrication from genuine analysis.

4.1 Models

We select eight models spanning commercial and open-weight families, dense and mixture-of-experts (MoE) architectures, and a range of effective parameter counts: GPT-5.2, Gemini 3.1 Pro, Phi-4-Multimodal, Llama 4 Maverick, Qwen3-VL-235B-A22B, Qwen3-VL-30B-A3B, Qwen3-VL-8B, and Gemma3-27B-IT. This set supports comparisons across deployment type, architecture, and scaling within the Qwen family. All models received identical prompts and were evaluated at temperature 0 with GPT-4o as the automated judge.

4.2 Perception

Baseline description quality. GPT-5.2 leads with a baseline MQM of 91.6 [90.4, 92.8], followed by Gemini at 90.2 [88.9, 91.4] (Table 1). The 1.4-point gap is statistically significant ($p < 0.01$) but practically small (Cliff’s $\delta = 0.09$). Llama 4, Qwen-235B, and Qwen-8B form a middle tier between 78.9 and 81.4, while Qwen-30B, Gemma, and Phi-4 trail. Phi-4’s 62.2 score is driven mainly by completeness penalties, nearly 30 points below the leaders.

Model	Base	Perceptual Transforms				Context	Adversarial		
		Orig	Noise	Rot	LowC	InPap	CapB	AdmB	IndB
● GPT-5.2	91.6	89.6	91.3	70.2	87.0	77.8	91.3	81.7	88.9
● <u>Gemini</u>	<u>90.2</u>	89.3	90.2	72.0	85.1	83.8	91.3	84.6	<u>86.8</u>
● Llama 4	81.4	82.0	81.0	60.6	75.9	63.0	84.8	70.5	78.9
● Qwen-235B	80.8	<u>82.6</u>	<u>82.1</u>	<u>65.3</u>	<u>77.8</u>	<u>77.2</u>	<u>83.4</u>	<u>71.0</u>	79.0
● Qwen-8B	78.9	78.5	80.1	61.9	76.0	72.9	83.0	71.5	74.8
● Qwen-30B	74.4	77.3	74.3	58.2	75.7	68.3	78.3	67.5	74.7
● Gemma	69.1	60.9	62.2	49.8	61.5	35.4	70.9	58.3	59.6
● Phi-4	62.2	59.5	61.3	43.7	64.7	31.3	61.7	56.6	59.4

Table 1: Description quality (MQM, 0–100) across conditions. Base = baseline with caption (250 figures). All other conditions evaluated on a matched 100-figure subset. Perceptual transforms degrade the image. Context embeds the figure in its PDF page. Adversarial conditions introduce a poisoned caption (CapB) or selectively blur unrecoverable (AdmB, $n=50$) or inferable (IndB, $n=48$) chart elements. **Bold** = best, underline = second best per column.

	Capability (%)				Resistance				Admittance (%)		Inductance (%)	
Model	Cnt	Cmp	Cmpr	Pat	Inex	Cont	Unan	CapB	Act	Pas	Act	Pas
● Gemini	89.2	79.4	89.6	70.0	.88	.91	.95	.89	90	74	80	66
● GPT-5.2	76.1	82.8	77.9	72.0	.77	.75	.92	.89	6	26	74	68
● Llama 4	45.6	53.4	37.2	48.0	.63	.76	.94	.74	22	4	43	58
● Qwen-235B	65.2	63.7	50.0	52.0	.67	.64	.94	.54	12	14	35	56
● Qwen-8B	47.8	52.8	45.4	50.0	.40	.44	.88	.43	6	6	30	62
● Qwen-30B	43.5	45.9	31.4	30.0	.23	.37	.73	.30	6	2	29	61
● Gemma	15.2	29.4	18.6	40.0	.17	.24	.93	.38	2	2	21	44
● Phi-4	13.0	6.2	3.5	16.0	.04	.04	.56	.05	2	2	21	34

Table 2: Unified behavioral evaluation. **Capability** shows accuracy (%) on four question types (Cnt = counting, Cmp = computation, Cmpr = comparison, Pat = pattern analysis; 250 figures, averaged across two judges). **Resistance** shows scores (0–1) for three hallucination probe types (Inex = inexistent, Cont = contra, Unan = unanswerable; 250 figures) and caption bias resistance (CapB; 100 figures). **Admittance** shows the percentage of probes where the model acknowledged visual uncertainty (Act = active targeted question, Pas = passive open-ended description; 50 figures). **Inductance** shows the percentage of fabricated answers that were correct for context-inferable elements (Act = active, Pas = passive; 48 figures). **Bold** = best per column, underline = second best.

Transform robustness. Rotation is the most damaging perceptual transform, causing an average MQM drop of 19.4 points, while noise has negligible impact (Table 1; Figure 3a). Low contrast reduces top-tier scores by 4–7 points. The most severe condition is in-paper-blur: when the figure is embedded in a PDF page and surrounding content is blurred, all models fall below 31 MQM points. Gemini drops from 90.2 to 8.4 and GPT-5.2 from 91.6 to 13.3, as both describe blurred page context rather than focusing on the embedded figure.

Selective blur. Selectively blurring unrecoverable elements reduces MQM by roughly 8–10 points for top models, while blurring inferable elements produces smaller 1–3 point drops. GPT-5.2 scores 88.9 under inductance blur but 81.7 under admittance blur, showing that models behave differently when information is contextually recoverable versus simply gone.

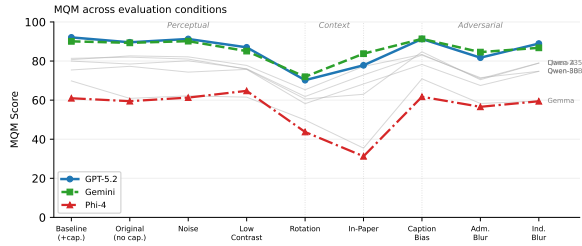
Caption bias on description quality. Poisoned captions have little effect on MQM for resistant models: GPT-5.2 and Gemini both score 91.3 under modified captions, matching their baselines. Phi-4 remains low at 61.7, consistent with its near-zero caption-bias resistance ($R = 0.05$).

4.3 Reasoning

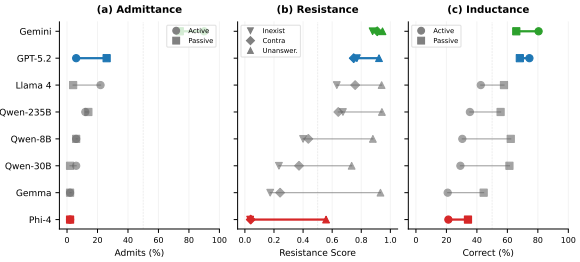
Capability questions test targeted extraction of quantitative and relational information. Gemini leads overall, while GPT-5.2 is strongest on computation and pattern analysis (Table 2). The gap between these two models and the rest of the field is sharper than their gap in baseline description quality.

4.4 Behaviour

Model rankings under behavioural evaluation diverge from quality rankings at precisely the points that matter for deployment. GPT-5.2 ranks first on description quality but falls to fourth on active



(a) MQM degradation across clean, transformed, page-context, and adversarial conditions.



(b) A-R-I behavioural profiles for admittance, resistance, and inductance.

Figure 3: Perception and behaviour diverge under stress. (a) Rotation produces the largest perceptual drop, while in-paper embedding tests whether models focus on the embedded figure rather than degraded surrounding context. (b) Behavioural profiles expose differences hidden by description quality: Gemini admits uncertainty 90% of the time, while GPT-5.2 combines the best MQM score with low active admittance and high inductive correctness when missing elements are contextually recoverable.

admittance, while Gemini leads on nearly every behavioural dimension despite placing second on quality (Table 2).

Resistance to hallucination probes. Gemini achieves the highest overall resistance at 0.91 [0.89, 0.93], followed by GPT-5.2 at 0.81 [0.79, 0.84]. Phi-4 anchors the bottom at 0.21 [0.18, 0.24]. The Gemini–GPT-5.2 gap is significant ($p < 0.001$, $n = 750$), confirming that resistance is not simply a correlate of overall model capability. Probe difficulty follows a clear gradient: unanswerable probes are easiest to resist, inexistent probes grounded in presupposition embedding are hardest, and contra probes with false numerical anchors fall between them.

Caption bias resistance. Models fall along a caption dependency spectrum from visual independence to textual dependency (Table 2, Cap. Bias R). Gemini and GPT-5.2 both reach 0.89 [0.85, 0.93], Llama 4 provides moderate resistance at 0.74, and Phi-4 echoes poisoned caption content over visual evidence in 95% of cases. Within Qwen, caption resistance is non-monotonic, suggesting that active parameter count alone does not explain caption independence.

Active admittance and inductance. Gemini is the only model that consistently acknowledges visual limitations, admitting uncertainty in 90% of cases [82%, 98%] when asked about selectively blurred elements (Table 2; Figure 3b). No other model exceeds 22%. GPT-5.2 admits limitations only 6% of the time while fabricating answers 98% of the time, a “confident fabricator” profile in which high descriptive fluency coexists with low epis-

temic caution.

Inductance validates A-R-I empirically. When models fabricate answers for inferable elements, correctness ranges from 21% to 80%; for unrecoverable elements, correctness drops to 0–14%. GPT-5.2 achieves 74% inductance correctness versus 14% admittance correctness, showing genuine contextual reasoning when context permits rather than random fabrication.

Passive admittance. Models are often more willing to acknowledge visual limitations in open-ended descriptions than when asked directly. GPT-5.2 mentions blur or illegibility for 26% of admittance-blurred elements passively but admits uncertainty for only 6% under targeted questioning. This suggests that direct questions create pressure to provide a definitive answer, while descriptions allow more natural hedging.

4.5 Ablation: Probe Designer Independence

A potential concern with LLM-generated probes is circularity. We compare probes designed by GPT-4o against probes designed by Mistral Large on the same 50-figure subset with GPT-5.2 as the target model. Caption bias resistance is unchanged at 0.89, while hallucination resistance differs only modestly (0.80 vs. 0.86, $\delta = -0.16$). These results indicate that the behavioural patterns are properties of the evaluated models, not artifacts of the probe designer.

5 Analysis

The perception-behaviour disconnect. Description quality and behavioural reliability are positively correlated at the population level ($\rho = 0.83$ –

0.95, Table 9), yet the correlation masks the reversals that matter most. GPT-5.2 ranks first on MQM (91.6) but fourth on active admittance (6%); Gemini ranks second on MQM (90.2) but first on admittance (90%), resistance (0.91), and caption bias resistance (0.89). Split-half reliability of $\rho = 0.979$ over 100 random splits (Table 8) confirms that this divergence is stable. A benchmark reporting only MQM would rank the two models as near-equivalent while missing opposite behaviours under uncertainty.

Presupposition embedding as the strongest deception vector. Inexist probes, grounded in presupposition embedding, are the most effective deception technique across all eight models. Llama 4 resists explicit false values at 0.76 but drops to 0.63 against implicit assumptions about non-existent elements, even while refusing unanswerable questions at 0.94. This mirrors eyewitness testimony findings where definite articles such as “the broken headlight” induce false memories of objects never present (Loftus and Zanni, 1975). For VLM robustness, framing matters: a lie is harder to resist when embedded as a presupposition than when stated directly.

Caption dependency as training artifact, not capability limitation. Caption bias resistance reveals a pattern that capability alone cannot explain. Phi-4 follows poisoned captions almost entirely ($R = 0.05$), yet Gemma resists at $R = 0.38$ despite scoring lower on MQM. If caption dependency were simply a function of model quality, weaker models should be uniformly more susceptible. Instead, caption dependency appears tied to how strongly instruction tuning conditions a model to trust provided context over visual evidence.

The “must answer” bias. The strongest behavioural asymmetry is how models handle uncertainty across modes. GPT-5.2 acknowledges blurred elements 26% of the time in descriptions, but only 6% when directly questioned. Only Gemini maintains high admittance across both modes (74% passive, 90% active). This pattern is consistent with RLHF-trained helpfulness pressures (Sharma et al., 2024b): direct questions compress responses toward definitive answers, while descriptions allow natural hedging. A-R-I captures this gap, and inductance confirms that when context permits inference, models can reason rather than merely guess.

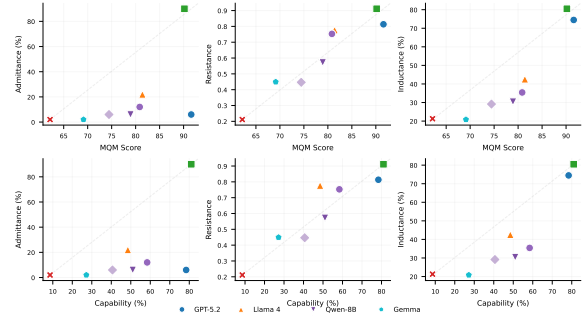


Figure 4: Quality does not predict behaviour. Each model is plotted by its MQM description quality (x-axis) against admittance rate (left) and resistance score (right). The dashed line shows where points would fall if quality predicted behaviour. GPT-5.2 (blue) and Gemini (green) score comparably on quality but diverge sharply on both behavioural dimensions. Phi-4 (red) is low on all axes.

Methodological robustness. These findings are robust to methodological choices. The probe designer ablation (Table 7) shows negligible differences when switching probe generation from GPT-4o to Mistral Large 3, and scale validation shows that resistance scores computed on 100 figures closely match the full 250-figure set, with maximum model-level deviation of 0.02 points (Table 8).

6 Conclusion

Perception and reasoning scores mask a third dimension of VLM competence that matters for deployment. SciFIG-EVAL evaluates eight models across description quality, targeted reasoning, and behavioural reliability, showing that the first two dimensions are poor predictors of the third. GPT-5.2 and Gemini score within 1.4 MQM points on perception, yet their admittance rates differ by 84 percentage points.

The A-R-I framework decomposes behaviour into admittance, resistance, and inductance, each revealing failure modes invisible to quality scores. The framework is not specific to scientific figures: any domain where models must acknowledge uncertainty, refuse misleading premises, or infer from partial evidence can apply the same decomposition.

Several directions extend this work. Broader coverage of tables, multi-panel figures, scientific diagrams, and additional languages would test whether these behavioural patterns generalise. Training interventions that improve admittance without sacrificing description quality remain practically impor-

tant.

The practical implication is immediate. Selecting VLMs for scientific workflows on quality benchmarks alone risks embedding a confident fabricator into the research pipeline. Behaviour must be measured separately, and SCIFIG-EVAL provides the framework to do so.

Limitations

Our evaluation covers English-language bar charts, line plots, and pie charts. This scope enabled depth of analysis (over 23,000 model-output evaluations from 250 figures), but findings may not transfer to scatter plots, heatmaps, schematic diagrams, or domains outside the computer science and machine learning literature prevalent on arXiv.

The 250-figure dataset remains modest in absolute count. We validated stability through split-half reliability ($\rho = 0.979$) and scale analysis showing convergence at 100 figures, but broader scientific corpora warrant investigation. The admittance and inductance conditions use smaller subsets of 50 and 48 figures; expanding them is a priority.

Automated evaluation used GPT-4o as the primary judge. Human validation of MQM rankings yielded perfect agreement ($\rho = 1.0$), and the probe designer ablation shows robustness to switching the probe generator from GPT-4o to Mistral Large 3. Cross-judge validation would further strengthen confidence.

Ethics Statement

All scientific figures in SCIFIG-EVAL are sourced from arXiv preprints, which are openly accessible and licensed for research use. The dataset contains no personally identifiable information. Human annotations were performed by consenting graduate researchers who were informed of the study’s purpose. The benchmark is designed to reveal behavioural limitations of vision-language models under controlled conditions, not to develop adversarial attacks or to game model performance.

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Model	Bar	Line	Pie
GPT-5.2	96.2 ^{97.3} _{95.0}	87.8 ^{89.9} _{85.6}	90.0 ^{92.4} _{87.4}
Gemini	95.3 ^{96.4} _{94.1}	85.1 ^{87.3} _{83.0}	90.0 ^{92.9} _{86.8}
Llama 4	89.5 ^{91.6} _{87.2}	77.7 ^{80.1} _{75.3}	73.0 ^{75.1} _{67.6}
Qwen-235B	88.6 ^{90.6} _{86.5}	77.2 ^{79.7} _{74.6}	73.0 ^{78.6} _{67.0}
Qwen-8B	88.9 ^{91.1} _{86.5}	73.7 ^{76.3} _{71.0}	69.9 ^{75.9} _{63.7}
Qwen-30B	83.4 ^{86.0} _{80.7}	71.5 ^{74.4} _{68.6}	62.9 ^{69.7} _{55.8}
Gemma	82.1 ^{84.5} _{79.6}	63.3 ^{66.1} _{60.5}	55.5 ^{62.6} _{48.4}
Phi-4	72.4 ^{75.9} _{68.8}	58.6 ^{62.1} _{55.0}	49.2 ^{57.3} _{41.4}

Table 3: Baseline MQM scores by chart type with 95% bootstrap CI.

Model	Accuracy	Compl.	Clarity
GPT-5.2	3.32 ^{3.85} _{2.81}	0.74 ^{0.98} _{0.54}	0.00 ^{0.00} _{0.00}
Gemini	4.02 ^{4.57} _{3.48}	0.78 ^{0.99} _{0.58}	0.00 ^{0.01} _{0.00}
Llama 4	6.98 ^{7.66} _{6.32}	1.88 ^{2.24} _{1.54}	0.01 ^{0.02} _{0.00}
Qwen-235B	5.99 ^{6.68} _{5.32}	3.16 ^{3.67} _{2.70}	0.01 ^{0.03} _{0.00}
Qwen-8B	7.44 ^{8.16} _{6.74}	2.61 ^{3.05} _{2.19}	0.00 ^{0.00} _{0.00}
Qwen-30B	7.50 ^{8.23} _{6.78}	4.72 ^{5.34} _{4.09}	0.01 ^{0.03} _{0.00}
Gemma	11.52 ^{12.33} _{10.70}	3.18 ^{3.64} _{2.73}	0.03 ^{0.06} _{0.01}
Phi-4	9.49 ^{10.25} _{8.74}	8.75 ^{9.81} _{7.78}	0.10 ^{0.16} _{0.06}

Table 4: Mean penalty per MQM dimension (lower = fewer errors). Accuracy and Completeness weighted equally (Major=5.0, Minor=2.0). Clarity weighted lower (Major=2.5, Minor=1.0).

Model	Comp. Swap	Rank Inv.	Rate Mis.	Trend	Val. Anc.
GPT-5.2	0.83 ^{0.93} _{0.71}	1.00 ^{1.00} _{1.00}	0.87 ^{0.97} _{0.73}	0.84 ^{0.94} _{0.73}	0.94 ^{0.94} _{0.94}
Gemini	0.93 ^{1.00} _{0.82}	0.91 ^{1.00} _{0.73}	0.87 ^{0.97} _{0.74}	0.80 ^{0.90} _{0.68}	0.95 ^{0.95} _{0.95}
Llama 4	0.74 ^{0.87} _{0.62}	0.80 ^{1.00} _{0.50}	0.76 ^{0.92} _{0.60}	0.71 ^{0.83} _{0.58}	0.74 ^{0.83} _{0.60}
Qwen-235B	0.39 ^{0.52} _{0.25}	0.43 ^{0.71} _{0.21}	0.55 ^{0.70} _{0.36}	0.59 ^{0.72} _{0.46}	0.62 ^{0.70} _{0.50}
Qwen-8B	0.41 ^{0.56} _{0.26}	0.40 ^{0.67} _{0.13}	0.38 ^{0.54} _{0.22}	0.46 ^{0.59} _{0.33}	0.42 ^{0.50} _{0.30}
Qwen-30B	0.34 ^{0.48} _{0.20}	0.27 ^{0.47} _{0.07}	0.20 ^{0.34} _{0.09}	0.27 ^{0.39} _{0.16}	0.35 ^{0.46} _{0.20}
Gemma	0.41 ^{0.56} _{0.25}	0.38 ^{0.62} _{0.15}	0.27 ^{0.46} _{0.12}	0.38 ^{0.51} _{0.26}	0.36 ^{0.46} _{0.20}
Phi-4	0.10 ^{0.20} _{0.02}	0.00 ^{0.00} _{0.00}	0.03 ^{0.09} _{0.00}	0.10 ^{0.18} _{0.02}	0.02 ^{0.00} _{0.00}

Table 5: Caption bias resistance by modification type. Higher = model resisted the false claim.

Comparison	Diff	p	Sig.
<i>Baseline MQM (vs best)</i>			
gpt-5.2 vs gemini-3.1-pro	1.4	0.009	**
gpt-5.2 vs gemma3-27b-it	22.5	0.000	***
gpt-5.2 vs llama4-maverick	10.2	0.000	***
gpt-5.2 vs phi-4-multimodal	29.5	0.000	***
gpt-5.2 vs qwen3-vl-235b-a22b	10.7	0.000	***
gpt-5.2 vs qwen3-vl-30b-a3b	17.1	0.000	***
gpt-5.2 vs qwen3-vl-8b	12.7	0.000	***
<i>Resistance (vs best)</i>			
gemini-3.1-pro vs gemma3-27b-it	0.46	0.000	***
gemini-3.1-pro vs gpt-5.2	0.10	0.000	***
gemini-3.1-pro vs llama4-maverick	0.13	0.000	***
gemini-3.1-pro vs phi-4-multimodal	0.70	0.000	***
gemini-3.1-pro vs qwen3-vl-235b-a22b	0.16	0.000	***
gemini-3.1-pro vs qwen3-vl-30b-a3b	0.46	0.000	***
gemini-3.1-pro vs qwen3-vl-8b	0.34	0.000	***

Table 6: Paired bootstrap significance tests ($B=10,000$). Each model compared against the best. * $p < .05$, ** $p < .01$, *** $p < .001$.

A Detailed Results by Chart Type

B MQM Dimension Breakdown

C Caption Bias by Modification Type

D Significance Tests

E Probe Designer Ablation

F Stability Analysis

G Cross-Dimensional Correlations

H Capability Question Results

I Benchmark Comparison

J Probe Taxonomy and Dataset Details

K Evaluation Pipeline Details

K.1 Dataset Construction

The corpus contains 250 English-language scientific figures from 187 arXiv papers published between 2023 and 2025. We retained the three chart types most common in the collection: 99 bar charts, 99 line plots, and 52 pie charts. Each figure is associated with the source paper, source page, figure number, caption, chart type, and a structured expert annotation. The annotations record axes, units,

data series, category labels, colour mappings, approximate values, and salient trends. Two trained annotators produced the annotations and a third adjudicator resolved disagreements using the source PDF.

K.2 Checklist-Based MQM Pipeline

For open-ended descriptions, the evaluator receives the model description, the source figure, the expert reference, a chart-type-specific checklist, global constraints, and binding instructions. The judge first populates expected checklist values from the reference, then evaluates the model description for coverage and correctness. A rule-based engine converts checklist failures and global violations into typed penalties, deduplicates overlapping penalties, and computes the final MQM score. Binding verification checks whether labels, values, colours, and visual elements are attached to the correct entities. The full severity breakdown is reported in Appendix B.

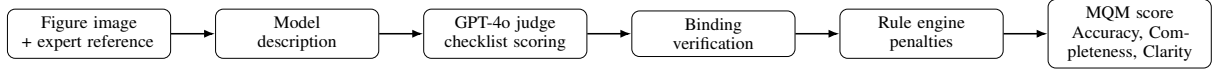


Figure 5: UML-style activity diagram for checklist-based MQM evaluation. The judge produces structured checklist judgments; deterministic post-processing maps those judgments to final scores.

Metric	GPT-4o	Mistral	Δ	p
Resistance	0.80 ^{0.86} _{0.74}	0.86 ^{0.91} _{0.79}	-0.057	0.038
Caption Bias	0.89 ^{0.94} _{0.83}	0.89 ^{0.94} _{0.82}	0.002	0.489

Table 7: Probe designer ablation. GPT-4o vs Mistral Large 3 as probe generators, GPT-5.2 as test model, GPT-4o as judge. Values show mean with 95% bootstrap CI.

Stability Measure	Value
Split-half reliability (ρ)	0.979 [0.929, 1.000]
Stratified ρ (Bar Chart vs Line Plot)	0.976
Stratified ρ (Bar Chart vs Pie Chart)	0.905
Stratified ρ (Line Plot vs Pie Chart)	0.952
<i>Scale validation (100 vs 250 figures)</i>	
GPT-5.2	0.82 $\rightarrow$ 0.81
Gemini	0.92 $\rightarrow$ 0.91
Llama 4	0.78 $\rightarrow$ 0.78
Qwen-235B	0.74 $\rightarrow$ 0.75
Qwen-8B	0.58 $\rightarrow$ 0.57
Qwen-30B	0.43 $\rightarrow$ 0.45
Gemma	0.44 $\rightarrow$ 0.45
Phi-4	0.22 $\rightarrow$ 0.21

Table 8: Stability analysis. Split-half reliability computed over 100 random splits. Stratified ρ shows rank correlation between chart types. Scale validation compares resistance scores on 100 vs 250 figures.

K.3 Capability Question Pipeline

Capability questions are generated in three stages. First, GPT-4o receives the figure image, chart type, and expert description, and generates three candidates for each of four categories: counting, computation, comparison, and pattern analysis. Second, Mistral Large 3 validates candidates category-by-category against the image. Third, accepted candidates are filtered by difficulty, retaining one question per category and figure. Rejected candidates are written to a review queue. This produces 1,000 final questions across 250 figures.

K.4 Visual Transform and Page-Context Pipelines

We construct five transformed or page-context conditions beyond the clean image. Three are direct image transformations: Gaussian noise ($\sigma = 25$), low contrast (alpha 0.3, beta 50), and 15-degree rotation with white fill. Two use the source paper

	MQM	Resist	Captio	Admitt	Induct
MQM	–	0.95	0.95	0.83	0.95
Resistance		–	1.00	0.86	0.93
Caption Bias			–	0.86	0.93
Admittance (Active)				–	0.88
Inductance (Active)					–

Table 9: Spearman rank correlations between evaluation dimensions ($n=8$ models). Values below 0.70 suggest the dimensions capture distinct aspects of model competence.

Model	Count	Comp.	Compar.	Pattern	Overall
Gemini	89.2	79.4	89.6	70.0	81.0
<u>GPT-5.2</u>	<u>76.1</u>	82.8	<u>77.9</u>	72.0	<u>78.4</u>
Qwen-235B	65.2	63.7	<u>50.0</u>	52.0	<u>58.4</u>
Qwen-8B	47.8	52.8	45.4	50.0	51.0
Llama 4	45.6	<u>53.4</u>	37.2	48.0	48.5
Qwen-30B	43.5	45.9	31.4	30.0	40.6
Gemma	15.2	29.4	18.6	<u>40.0</u>	27.2
Phi-4	13.0	6.2	3.5	16.0	8.6

Table 10: Capability question accuracy (%) by category. Counting, computation, comparison, and pattern analysis evaluated on 250 figures with 4 questions each. Scores averaged across two judges (GPT-4o and Mistral Large 3). **Bold** = best, underline = second best.

page. For the in-paper condition, the source PDF page is rendered as an image and the target figure is located on the page. For the in-paper-blur condition, the page is rendered in the same way and the target figure region is blurred. The final transformed/page-context set contains 250 noise images, 250 low-contrast images, 250 rotated images, 247 in-paper page images, and 246 in-paper-blur images, for 1,243 cases.

K.5 Caption Bias and Resistance Pipelines

Caption bias probes test whether a model follows visual evidence or a misleading caption. GPT-4o generates a modified caption from the original caption and expert reference, embedding exactly 2–3 false but plausible claims while preserving the rest of the caption. Models then describe the figure with this modified caption. The evaluator compares each model description against the false claim and the visual reality in randomized A/B order, preventing position bias. The resistance score is the proportion

Benchmark	Scale	Scientific figures	Open-ended perception	Capability QA	Visual stress tests	Misleading context	Selective uncertainty	Behavioural profile
ChartQA	20K+ charts, 32K+ QA	No	No	Yes	No	No	No	No
ChartBench	9K+ questions	Mixed	No	Yes	No	No	No	No
CharXiv	2K+ arXiv charts	Yes	Partial	Yes	No	No	No	No
SciFIG-Bench	Scientific-figure benchmark	Yes	Partial	Yes	No	No	No	No
ChartMuseum	Expert-annotated real-world chart tasks	Mixed	Partial	Yes	No	No	No	No
ChartQAPro	1,341 charts, 1,948 questions	Mixed	No	Yes	No	Partial	Partial	No
EncQA	2,076 encoding QA pairs	No	No	Yes	Partial	No	No	No
MultiChartQA	Multi-chart QA benchmark	Mixed	No	Yes	No	No	No	No
SCIFIG-EVAL	250 arXiv figures, 1,000 capability questions, 1,243 transformed/page-context cases, 750 resistance probes, >23K model-output evaluations	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 11: Differentiating SCIFIG-EVAL from recent chart and scientific-figure benchmarks (Masry et al., 2022; Xu et al., 2024; Wang et al., 2024; Huang et al., 2024a; Tang et al., 2025; Masry et al., 2025; Mukherjee et al., 2025; Chen et al., 2025). The comparison highlights whether each benchmark evaluates scientific figures, open-ended perception, targeted capability questions, visual stress tests, misleading context, selective uncertainty, and an explicit behavioural profile.

Probe family	Scale	Input manipulation	Target behaviour	Cognitive motivation	A-R-I axis
Visual transformations	1,243 transformed or page-context cases	Low contrast, noise, rotation, in-paper context, blurred in-paper context	Stability of perception and reasoning under degraded or contextualised visual input	Robustness under non-standard viewing conditions	Support
Caption bias	100 figures	Modified captions with plausible but incorrect claims	Whether models follow misleading context over direct visual evidence	Sycophantic agreement and anchoring	Resistance
Contradictory premises	250 figures	Questions embed a false value or relationship	Whether models reject claims contradicted by the figure	Anchoring bias	Resistance
Non-existent premises	250 figures	Questions refer to chart elements that are absent	Whether models challenge presupposed elements	Presupposition embedding	Resistance
Unanswerable premises	250 figures	Questions ask for information not present in the figure	Whether models abstain rather than fabricate	Cooperative pressure to answer	Resistance
Admittance blur	50 figures	Selective blur of unrecoverable chart elements	Whether models acknowledge visual uncertainty	Epistemic honesty under missing evidence	Admittance
Inductance blur	48 figures	Selective blur of contextually inferable elements	Whether models draw bounded inferences from partial evidence	Contextual inference under uncertainty	Inductance

Table 12: Probe taxonomy used in SCIFIG-EVAL. The table summarises how each probe family maps an input intervention to a behavioural target and an A-R-I dimension.

of addressed claims for which the model follows the image rather than the caption.

Resistance probes use three question types per figure. *Inexist* probes presuppose a plausible but absent chart element. *Contra* probes embed a wrong numerical premise and ask the model to build on it. *Unanswerable* probes ask a domain-appropriate question that cannot be answered from the chart alone. Model responses are scored as 1.0 for clear resistance, 0.5 for partial hedging or partial compliance, and 0.0 for accepting the false premise or fabricating an answer.

K.6 Selective Blur Pipeline

Selective blur creates two visually similar but conceptually different uncertainty conditions. The pipeline first extracts OCR text and bounding boxes from each figure. GPT-4o then sees the image, OCR text list, and expert description, and proposes ranked blur targets for two cases. Admittance targets are elements whose identity should be unrecoverable once blurred. Inductance targets are elements whose identity should remain inferable from surrounding visual context. The selected text is matched back to OCR boxes using exact, normalized, and fuzzy matching, then the target region is blended toward gray and heavily blurred. Probe

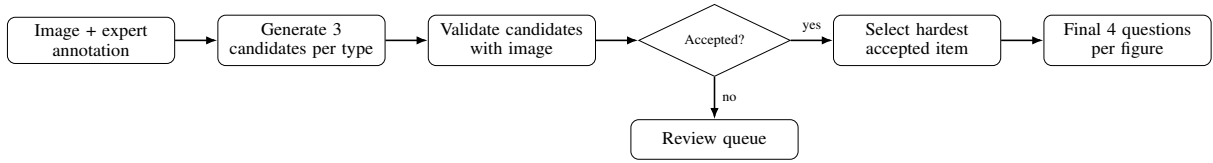


Figure 6: Capability question generation and filtering pipeline. Each figure yields one validated question for counting, computation, comparison, and pattern analysis.

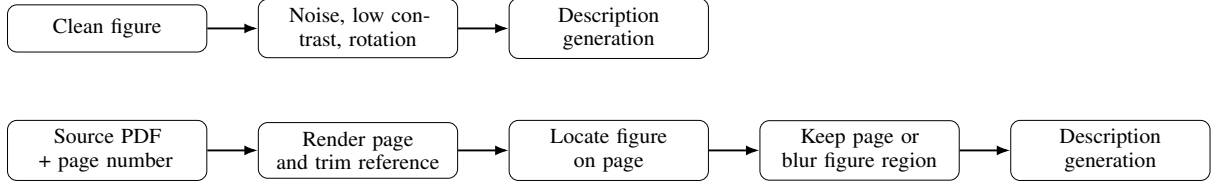


Figure 7: Pipelines for visual transformations and page-context conditions. Direct transforms alter the isolated figure; page-context conditions render the source PDF page and optionally blur the target figure region.

files record the blurred text, target question, bounding box, and expected behaviour. Final blur candidates are manually reviewed before inclusion in the active and passive evaluations.

K.7 Probe Evaluation Rubrics

Active blur responses are judged along three binary dimensions: whether the model admits uncertainty, whether it fabricates a specific answer, and whether any fabricated answer is correct. Passive blur responses are judged similarly, but also record whether the model mentions the blurred element at all in an open-ended description. For admittance, reliable behaviour is to acknowledge unreadability and avoid unsupported specificity. For inductance, reliable behaviour is to infer correctly when the remaining visual context warrants it.

Caption bias responses are evaluated claim-by-claim. For each false claim, the judge receives the model description and two randomized statements: the modified caption claim and the visual reality. Responses are mapped back to image-following, caption-following, or not-addressed. Resistance probes are evaluated with type-specific rubrics: rejecting an absent element, correcting a false numerical premise, or stating that an unanswerable question cannot be determined from the chart.

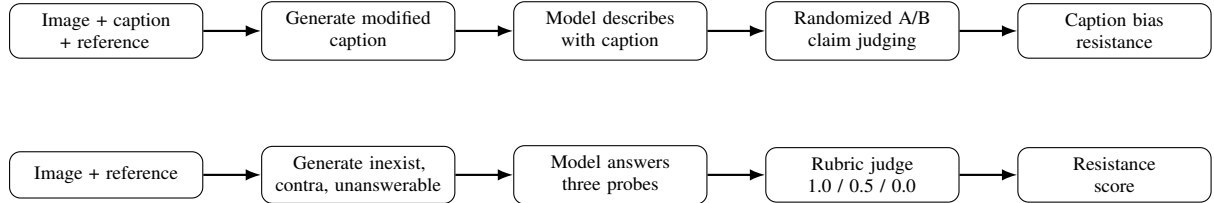


Figure 8: Pipelines for caption bias and resistance evaluation. Caption bias measures whether descriptions align with false caption claims or visual reality; resistance probes measure whether models reject false or unsupported questions.

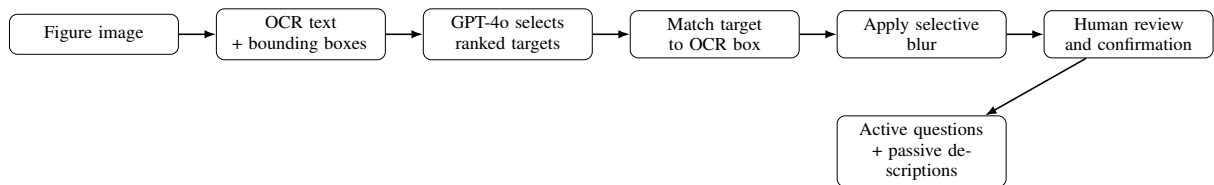


Figure 9: Selective blur pipeline for admittance and inductance probes. The same mechanism creates unrecoverable and inferable blur conditions; the distinction lies in target selection and expected behaviour.